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A Voting Ensemble Learning Model for Improved Credit Default Risk Prediction

Parambrata Sanyal
Symbiosis Institute of Technology, Nagpur Campus, Symbiosis International (Deemed University),
Pune, India
sparambrata26@gmail.com

Mukund Kuthe
Symbiosis Institute of Technology, Nagpur Campus, Symbiosis International (Deemed University),
Pune, India
mukundk0204@gmail.com

Sudhanshu Maurya
Symbiosis Institute of Technology, Nagpur Campus, Symbiosis International (Deemed University),
Pune, India
dr.sm0302@gmail.com

Kashish Mirza
Department of Computer Science & Engineering,
Graphic Era Hill University, Bhimtal Campus, India
kashishmirza99@gmail.com

Pradnya Borkar
Symbiosis Institute of Technology, Nagpur Campus, Symbiosis International (Deemed University),
Pune, India
pradnyaborkar2@gmail.com

Rachit Garg
School of Computer Science and Engineering,
Lovely Professional University, Jalandhar,
India
rachit.garg.in9@gmail.com

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***Abstract*— Credit risk prediction is one of the most rudimentary pillars for practices in financial institutions. In order to maintain sound financial health and support the long-term safety of the institution, a strong framework for credit default risk evaluation is necessary. A strong credit default risk forecasting framework ensures financial institutions expand their lending activities and explore new markets. In different cities across the world, loan defaults in financial institutions result in adverse impacts such as financial losses, reduced profitability, and capital adequacy ratios. Therefore, it is high time to overcome traditional credit evaluation methods that are being followed in several financial institutions till date. This paper thoroughly evaluates the loan default prediction methodologies or similar assessment techniques proposed till date. Exploring all the best possible machine learning (ML) techniques, this paper proposes a soft-voting ensemble model combining three different (Random Forest, XG Boost, Gradient Boosting) and high-performing models to forecast credit default risks ensuring safety and success in institutional lending activities, evaluating creditworthiness of the borrower. The dataset was acquired from Kaggle. Following data preprocessing, the processed data was trained into the proposed ensemble model, followed by hyperparameter tuning, evaluation, and deployment of the model. The evaluation metrics include the F1 score, recall rate, accuracy, AUC-ROC and model precision. The final data visualizations portray the comparisons and the benefits of using this model across different financial institutions.**

Keywords*— *Machine Learning (ML), Credit Default Risk, Ensemble Classifier, Soft-Voting Ensemble, Random Forest (RF), XG Boost, Gradient Boosting (GB).

I. Introduction

Credit Risk is an occurrence, or potential for financial losses that occurs due to a borrower's failure to repay the amount of debt and meets contractual obligations. Generally, credit risks are faced by the ones who extend credit, such as financial institutions, investors, and businesses.

With the ability to deal with a large number of datasets, make complex decisions, and be adaptive, Machine Learning models have become significant in this predicting role. With efficient handling and good methodology, financial institutions can migrate these risk issues, make improved judgments and reduce their overall loss. Unlike artificial intelligence, machine learning models are non-biased as they rely on objective data and algorithms. These algorithms allow proactive prediction due to pattern analysis techniques. They can be deployed to real-time scenarios in order to make timely decisions. By analyzing large datasets, machine learning algorithms can extract meaningful and exact insights and then provide judgments.

Figure 1 below displays the different components of Credit Risks. System Risk refers to the widespread financial crisis adversely impacting credit markets and institutions; Market Risk refers to the loss in a particular market due to fluctuations in commodity prices, exchange rates, and interest rates. Enterprise Risk underscores various risks, including credit-related legal issues, institutional credit reputation risk, and operational risk. Funding Risk is when the institution fails to obtain its credit requirements, potentially leading to liquidity problems. This paper focuses on Default Risk, which refers to the potential loss of lenders resulting in a borrower's inability to return their debt.

Fig 1. Different Components of Credit Risks

A. Motivation:

In the last few years since the Covid-19 pandemic, countries worldwide have witnessed several credit defaults. According to S & P Global reports, “2020 saw the highest number of credit defaults since 2009”. Furthermore, countries like Sri Lanka and Zambia have witnessed sovereign credit defaults in the year 2022. The S & P Global report shows that the sharp increase in credit defaults nearly doubled in the last two years in Italy. CRISIL reports a slight increase in credit defaults in 2023 in India. Proskauer’s reports show the continuation of the rise of private credit defaults by 2.71% in 2024 (New York). These are some of the minor data taken as a reference. This impacts the overall countries’ economy and affects institutional lending activities.

B. Model Flow:

The proposed methodology involves the sequential flow of data acquisition, model training and tuning, and model evaluation and deployment. The evaluation section of the paper involves a thorough evaluation of different machine-learning models along with the proposed voting ensemble model. The basic flow diagram of the proposed methodology is shown in Figure 2 below.

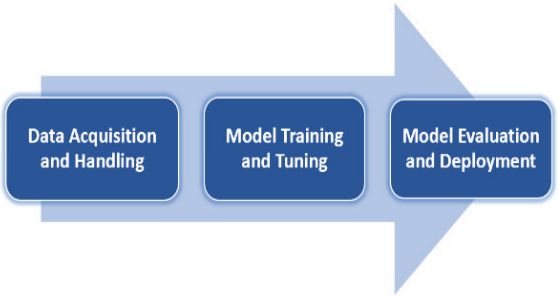

Fig 2. Process Flow Diagram of Proposed Methodology

- The initial steps involve relevant data acquired from trusted sources (Kaggle data in this paper) providing the borrower’s historical data. This data undergoes different stages such as handling missing values, removing outliers, cleaning, scaling, splitting of data into training and testing the dataset
- The dataset has been trained in different machine-learning models, including the proposed ensemble model. Further tuning the model in order to enhance its performance.
- The last stage involves model evaluation using appropriate metrics to assess efficiency and performance. If results are praiseworthy enough for real-world deployment, they can be deployed.

The entire process flow has been explained in the Proposed Methodology section of this paper.

II. Review Literature

In the recent past, different authors from across different parts of the globe have developed their methodologies to forecast and evaluate credit default risks. Based on various techniques and different evaluation metrics, this problem has been approached, hereby presenting the past works.

In their paper, N. Robinson and N. Sindhwani use cutting-edge machine learning techniques such as Gradient Boosting and random Forest classifiers on a processed dataset to enhance credit risk assessment in financial institutions. The model was evaluated using accuracy and precision [1]. Y. Ouyang et al., to forecast loan default risk, employed both Logistic Regression and XG Boost techniques. The prediction accuracy of the XG Boost model surpasses that of Logistic Regression following preprocessing and training [2]. A comparison between the Decision Tree classifier, Gaussian Naïve Bayes, and Linear Regression concludes that despite drawbacks, the Decision Tree classifier is useful for loan approval in financial institutions, enhancing its readability and clarity [3]. The research work of J. Hedrick et al. evaluates six different machine learning models, including Random Forest, Decision Tree, Support Vector Machines, Naïve Bayes, Artificial Neural Networks, Multi-Layer perceptron and a Stacking ensemble model, which was trained using a dataset with 20 different factors. As a result, after evaluation, the Stacking ensemble model comes up with the highest accuracy and precision, and only Random Forest comes close to that [4]. M. Z. Hussain et al. highlights that with the rise in loan applications in the recent past how it is difficult for financial institutions to make wise lending judgments. The paper comes up with a rigorous model evaluation of Random Forest, Naïve Bayes, stacking model, and Multilayer perceptron, clearly surpassing the traditional methods of credit score evaluation in institutions [5]. B. Gao et al., in the paper, validate two techniques, KNN and BP neural network, for loan default prediction. With the BP neural network model surpassing all evaluation metrics, the paper claims that proactive risk migration is possible with the model’s early forecasting of potential default hazards [6]. Y. Zhang shows how the Linear prediction approach can accurately focus marketing outcomes. The technique basically performs data processing by K means clustering algorithm [7].

Artificial intelligence, along with machine learning, is significant in the efficient evaluation of borrowers’ creditworthiness through improved insights into clients’ financial behavior. Using AI and ML can improve accuracy in this automated process, which is probably the most popular way of evaluation in the near future [8]. J. J. Alexander and H. Tjahyadi took Indonesian Regional Banks as reference data, and the study focuses on and assesses credit default prediction using machine learning. The results point to Naïve Bayes as a noteworthy technique for managing credit risk in local banks, offering early default detection and enhanced financial stability [9]. Based on logistic regression and Bernoulli distribution, credit risk prediction solves optimal loan allocation problems, provides a loan interest choice model, forecasts the probability of default, and uses key indicator data from raw invoices [10]. The author [11] highlighted the high accuracy of the Random Forest algorithm, resulting in the lowering of credit default risks [11]. M. N. Alam and M. M. Ali stated the limitations of ML models without semantic data; therefore, they have proposed knowledge graph theory in credit default prediction [12]. Y. Xiao et al. explained China’s Small and Medium-sized businesses (SMEs) as a difficult area and proposed loan allocation techniques to them despite their low financial strength and underperforming creditworthiness. The paper uses the Ant-Colony Algorithm to accurately forecast SMEs' credit defaults [13]. U. E. Orji et al. evaluate different machine learning algorithms on historical datasets; Random Forest comes up with the highest score and Linear Regression the lowest after evaluation. However, all the machine learning techniques surpass the traditional loan approval model [14].

S. Barua et al. [15] used a document verification module, a personalized module for accurate loan default forecasting, and implemented the Cat Boost algorithm. In comparison with Random First and Gradient Boosting methods, Cat Boost is found to surpass RF and GB in all metrics [15]. Y. Xie, et al. [16] have taken China's agricultural credits as a problem area. With over 500 million people living in rural areas, China's agriculture sector has difficulties in mitigating the risk associated with rural loans. At this moment, business professionals' experience is used to anticipate credit risk. In order to estimate credit risk in Chinese farmers, this research suggests a novel deep neural network model called DNN-CRP, which performs better than existing models [16].

In online credit lending services, fraud detection and forecasting play a critical role in managing loss risk and enhancing effectiveness. (H. Zhu and C. Wang) by reducing information carriers and enhancing valuable information, presented a unique multi-stage data representation system called AI2Vec (Applicant Information Vectoring) to improve information density. By using this method, machine learning classifiers can surpass the most advanced techniques [17]. The related works done to date demand a powerful ensemble model with proper data handling and model tuning. Our papers aim to fulfill that by proposing a soft voting classifier-based ensemble model that advances the existing models and provides a more accurate and efficient tool for default predictions. This paper evaluates five machine learning-based models (RF, XG Boost, GB, Decision Tree, Gaussian Naïve Bayes, and Proposed Ensemble Model).

a) Random Forest: The random forest is a powerful ensemble learning methodology that makes predictions by incorporating several decision trees. A random forest works by making a forest with lots of decision trees, with each tree trained on different parts of the data. Each and every one of the forest trees votes in prediction, and then it chooses the most frequent prediction to set the final result. This helps in reducing the overfitting; hence, the overall performance of the model is improved.

b) Decision Tree: Decision Trees are machine learning models that make decisions based on a series of questions posed in an 'if-else' fashion. A tree-like model, where each node is a decision, every branch is a possible outcome, and each leaf encompasses a prediction. Decision trees are straightforward for humans to comprehend and interpret; however, they can be prone to overfitting.

c) XGBoost: In fact, this is the acronym for (eXtreme Gradient Boosting), showing one of the most powerful algorithms in machine learning. The popularity of this algorithm can be explained by its high speed, accuracy for a wide range of predictiveness, and handling big volumes of data. XGBoost uses a regularized gradient-boosting framework that prevents overfitting and increases generalization performance.

d) Gradient Boosting: This is another ensemble method which involves training models iteratively to correct errors from previous models. It works by continuously adding new models to the ensemble while focusing on the mistakes earlier models have made. Overall, gradient boosting is a very powerful technique that can be used either for classification or regression.

e) Naive Bayes: This is a probability-based classifier, a classifier based on the Bayes theorem. That is, with regard to a certain data point, all of its features are not dependent on each other, given its class label. It is a simple and straightforward algorithm that has broad applications in the classification of text and spam filtering.

III. Proposed Model

A pipeline that started with raw data and passed to pre-processing predominantly through missing value handling, outlier removal, and cleaning, as well as scaling, is shown in figure 3. The result was divided into 80 percent of the training data and 20 percent of the data for testing. With the strength of the collaborative multiple machine learning algorithms, namely RF XGB and GB, the Voting Ensemble Model is improved for the selection of accuracy in identifying diversity.

A. *Data Acquisition and Handling:*

The data has been collected from Kaggle in a CSV file. Table I explains the dataset. The dataset comprises 4000 instances and over 10 features, each instance for a different loan ID.

TABLE I. Data Description Table

S.No.	Feature	Type of Data
1	Loan Identity of Borrower	Integer Type
2	Number of Dependents	Integer Type
3	Annual Income	Integer Type
4	Amount of Loan Required	Integer Type
5	Term of Loan	Integer Type
6	Cibil Score	Integer Type
7	Value of Residential Assets	Integer Type
8	Value of Commercial Assets	Integer Type
9	Value of Luxury Assets	Integer Type
10	Value of Bank Assets	Integer Type

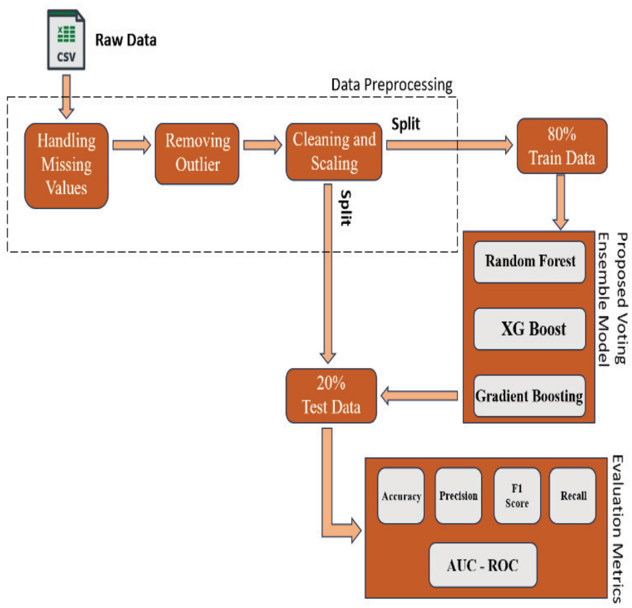

Fig 3. Proposed Model Architecture

a) Data Preprocessing: The dataset has been cleaned, outliers were removed, and missing values were handled before analysis. Ensuring the quality and dependability of the data used for model training is crucial at this stage.

b) Data Preprocessing: The dataset has been cleaned, outliers were removed, and missing values were handled before analysis. Ensuring the quality and dependability of the data used for model training is crucial at this stage.

c) Data Splitting: After cleaning, the data is separated into subgroups for testing and training. This split allows one portion of the data to be used for training machine learning models and holds another portion in order to evaluate the capability of the model. The train and test ratio was 3:1 (75%, 25%), and a random state variable, 42, was chosen.

d) Feature Selection: It selects important features that are relevant for the prediction of credit default risk. Feature selection includes various steps involving statistical analysis and domain knowledge to identify which features are useful in providing the predictive power of a model.

From this viewpoint, data acquisition forms the basis on which the credit default risk prediction model stands. Thus, the quality and relevance of such data directly affect the accuracy and effectiveness of predictions made by machine learning algorithms.

Moderate positive correlations are found between

- “Annual Income” vs “Residential Assets Value”
- “Annual Income” vs “Commercial Assets Value”
- “Loan Amount” vs “Residential Assets Value”
- “Loan Amount” vs “Commercial Assets Value”

Heatmap showing the Correlation between features is depicted in Figure 4.

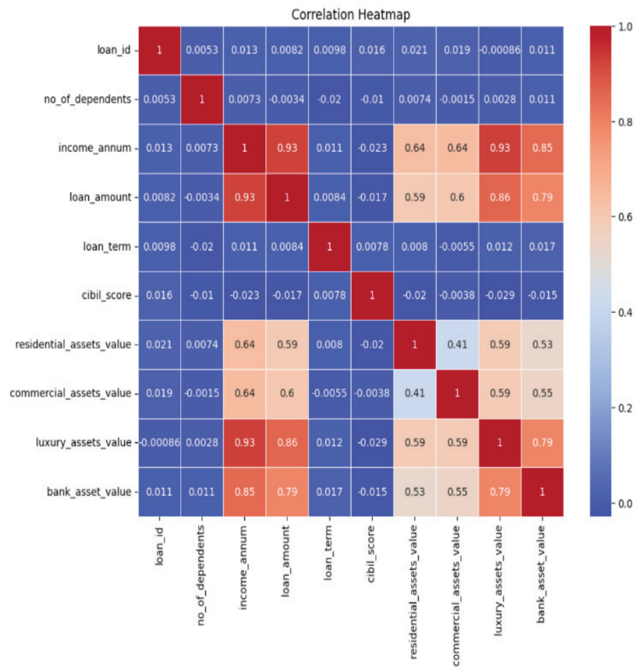

Fig 4. Correlation between Features

B. *Evaluation Metrics:*

In order to efficiently evaluate the performance of the machine learning models, the following evaluation metrics are chosen. Key Terms that are involved:

TP: True Positives, TN: True Negatives, FP: False Positives, FN: False Negatives

- Accuracy is defined by the proportion of correct predictions, i.e., True Positives and True Negatives, out of the total number of predictions.

Accuracy = (TP + TN) / (TP + FP + TN + FN)

- Precision is the proportion of True Positive predictions to positive predictions.

Precision = (TP) / (TP + FP)

- Recall measures the sensitivity of models in positive instances. High recall indicates it is less likely to miss positive cases.

Recall = (TP) / (TP + FN)

- F1 Score is balanced considering both precision and recall. It is a harmonic mean of both precision and recall.

F1 = 2 * (Precision * Recall) / (Precision + Recall)

- AUC-ROC is the measure of the model’s ability to distinguish between positive and negative classes.

All the above metrics depend upon the quality of the dataset in order for the correct evaluation of the models.

IV. RESULTS AND DISCUSSIONS

The Ensemble model leads in all evaluation metrics. However, XG Boost comes close in all aspects, followed by Gradient Boosting and Random Forest. The tables II shows the results of the models based on each evaluation metric.

TABLE II. ACCURACY AND PRECISION

Model	Accuracy	Precision
Ensemble Model	0.862	0.857
XG Boost	0.851	0.848
RF	0.823	0.815
GB	0.837	0.829
Decision Trees	0.789	0.772
Naïve Bayes	0.754	0.731

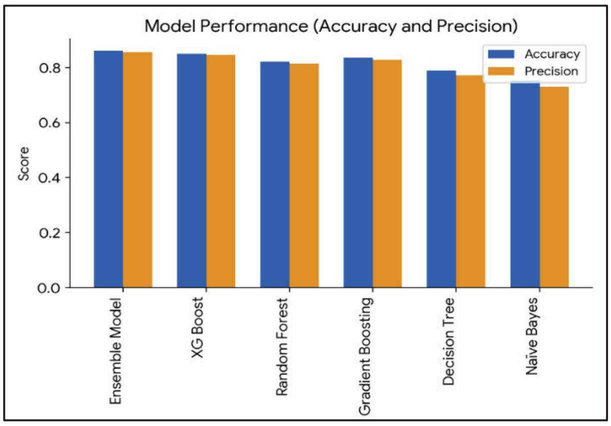

Fig 5. Column Graph Showing Accuracy and Precision Data

TABLE III. RECALL RATE AND F1 SCORE

Model	Recall Rate	F1 Score
Ensemble Model	0.871	0.864
XG Boost	0.862	0.855
RF	0.831	0.823
GB	0.845	0.837
Decision Trees	0.807	0.789
Naïve Bayes	0.778	0.754

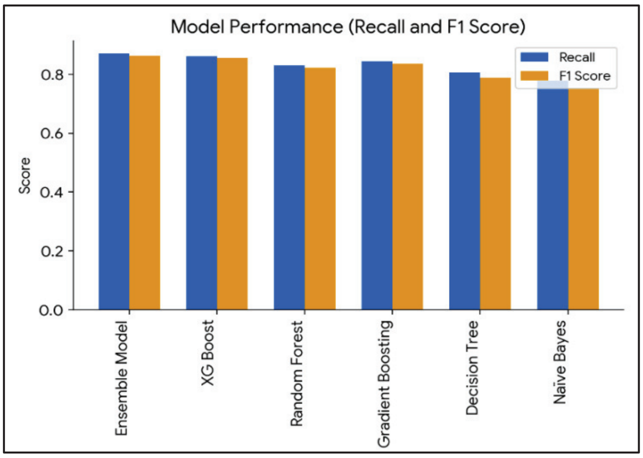

Fig 6. Column Graph Showing Recall and F1 Score

A. *Model Performance:*

The ensemble Model performs the best among all the model's overall metrics. It scores an accuracy of 0.862 and a precision of 0.857. XGB is very close to this model with an accuracy of 0.851 and precision of 0.848, thus proving good credit risk predictability. Other models in consideration are RF and GB, which show competitive results and are not comparable to that of the ensemble model. This ensemble model is able to get the recall rate of 0.871 and the F1 score of 0.864 so that it identifies positive instances, which are actual defaults.

TABLE IV. AUC-ROC

Model	Auc-Roc
Ensemble Model	0.913
XG Boost	0.901
Random Forest	0.872
Gradient Boosting	0.884

All the visualizations (Figure 5,6,7) depict the superiority of the proposed Ensemble Model over all the machine learning models in all the metrics, including accuracy, AUC-ROC, precision, recall rate, and F1 score. The study findings conclude the robustness of triple voting ensemble model in predicting Loan Defaults. Moreover, the data chosen is unbiased with respect to Age, Gender, and Region.

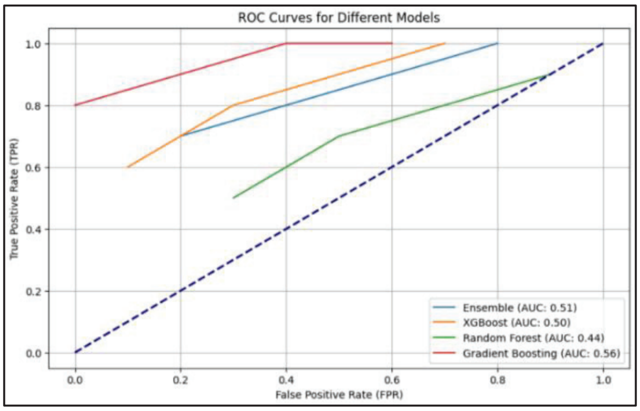

Fig 7. ROC Curve of Four Different Models

V. CONCLUSION

The proposed soft-voting ensemble model, which uses a combination of RF, XG Boost, and GB, gives much better credit default risk precision than existing works. This has established that many machine learning techniques combined in a model through an ensemble method tend to give improved metrics of F1 score, recall rate, accuracy, AUC-ROC, and precision. This would be of paramount importance for the financial institutions that intend to proceed with prudent lending decisions and avoid the risks of loan defaults. The paper has not only highlighted the effectiveness of the proposed ensemble model but also opened avenues for further consideration and enhancement of methodologies for credit risk prediction to keep them relevant and effective in an ever-changing financial world.

VI. FUTURE WORKS

While ensembles generally reduce overfitting, boosting models such as XGB and GB can still suffer from overfitting conditions, especially for smaller datasets, if not properly set. Mismanaging their hyperparameters, including learning rate, tree depth, and number of estimators, may cause such a model to overfit, thereby failing to generalize very well to unseen data. The future direction of this work may be toward incorporating all other data sources, such as social network behaviors and transaction data, to provide a more robust credit risk model.

Further scope includes deep learning and reinforcement learning algorithms for better performance. In this way, financial organizations can perform lending decisions immediately with real-time prediction models. Moreover, the methodologies could be adopted for industries such as insurance and investment, which would extend the range of their usefulness. This is followed by the recommendation for longitudinal studies assessing the long-term efficacy of the proposed model during different economic conditions. Addressing ethical considerations is finally the way in which algorithmic bias can make sure that credit risk predictions are nondiscriminatory against certain groups of borrowers.

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